

In Brightest Day, In Micro-Watts: A Feasibility Framework for Wearable Multimodal Acoustic-Optical Force Fields Driven by Neural Intent

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Abstract—For over eight decades, comic literature has asserted that an emerald ring fueled by raw, unyielding “willpower” can instantaneously materialize tangible green locomotives, defensive shields, and oversized boxing gloves in mid-air. In terrestrial physics, however, continuous radiation pressure from light capable of arresting a falling vehicle demands several terawatts of power, instantaneously vaporizing targets while converting the wearer’s metacarpals into dust via Newton’s third law. In this paper, we ground the legendary Green Lantern Power Ring into known physical mechanics. We propose a finger-worn form factor combining a 20×20 mm piezoelectric micromachined ultrasonic transducer (PMUT) phased array ($f_0 = 170$ kHz) interleaved with 532 nm green vertical-cavity surface-emitting lasers (VCSELs). By substituting comic-book hard light with airborne Gor’kov potential wells, the array synthesizes closed acoustic “twin-traps” capable of levitating tracer particles into persistent volumetric shapes while imparting localized tactile radiation pressure onto skin. Driven by a non-invasive Brain-Computer Interface (BCI) with an end-to-end latency budget of 24.11 ms, this architecture provides a biologically plausible and physically bounded prototype for thought-driven free-space constructs—all without requiring an extraterrestrial lithium-ion lantern on Oa.

Index Terms—Acoustic levitation, Brain-Computer Interface (BCI), Gor’kov potential, MEMS transducers, multimodal displays, radiation pressure, wearable computing.

I. INTRODUCTION

FOR over eighty years, DC Comics has presented a remarkably consistent engineering claim: slipped onto an index finger, an alien ring fueled by the “emotional spectrum” of willpower can overcome physical impediments, synthesize complex machinery out of green light, and enforce galactic law across Sector 2814 [1]. Whether manifesting a giant glowing flyswatter to neutralize an interstellar invader or throwing up a geodesic dome to catch falling debris, the Power Ring remains the gold standard of wearable human-machine interfaces.

Regrettably, standard terrestrial physics is notoriously stubborn regarding comic-book wish fulfillment. When Hal Jordan catches an oncoming 1,500 kg automobile using a “hard-light” force field, he is manipulating electromagnetic momentum. To exert the 14.7 kN of vertical force required to halt such a mass using pure optical radiation pressure from a perfectly reflecting

construct ($R = 1$), the necessary continuous beam power P is dictated by:

$$P = \frac{F \cdot c}{2} \approx 2.21 \times 10^{12} \text{ W} = 2.21 \text{ TW} \quad (1)$$

where $c \approx 3 \times 10^8$ m/s is the speed of light in vacuum.

Equation (1) yields an instantaneous power requirement exceeding the electrical generation capacity of the entire United States power grid. Worse still, if the underside of the caught automobile absorbs even 0.1% of this optical energy ($\alpha = 0.001$), the dissipated thermal load (2.21 GW) would vaporize the chassis and its occupants into ionized plasma within milliseconds. Simultaneously, conservation of momentum dictates that an equal and opposite reaction force of 1.5 metric tons would drive directly back down into the operator’s finger, obliterating their skeletal architecture before the first heroic quip could be delivered.

If a Power Ring cannot rely on petawatt lasers, nor on relativistic pair-production matter synthesis ($E = mc^2$, where a modest 1 kg titanium-equivalent shield commands an explosive budget equivalent to a 21.5 megaton thermonuclear yield), must the concept remain relegated to comic pages?

Not necessarily. The functional illusion of a “hard-light construct” requires only two physical ingredients:

- 1) **Visual Confinement:** The human eye must perceive a 3D geometry floating in mid-air (preferably glowing in the 520–535 nm emerald spectrum).
- 2) **Tactile Resistance:** The human somatosensory system must register mechanical impedance when interacting with the volume, tricking Mechanoreceptor Type I and II units into experiencing a “solid” boundary.

Recent advances in volumetric displays demonstrate that airborne acoustic phased arrays can create micro-Newton to milli-Newton radiation force fields in open space [3], while simultaneously trapping lightweight tracer particles via acoustic radiation potentials to project volumetric figures via persistence of vision [4]. Concurrently, wearable Brain-Computer Interfaces (BCIs) and high-density surface electromyography (sEMG) have matured to decode motor intent and directional vectors well within the cognitive sensorimotor loop [5].

In this paper, we bridge the gap between speculative comic lore and applied physics by formulating the architectural, mathematical, and latency limits of a real-world, finger-worn

construct generator. The primary contributions of this work are:

- A rigorous baseline analysis demonstrating why optical radiation pressure fails as a wearable macroscopic structural barrier under relativistic and thermodynamic constraints.
- A design blueprint for a 20×20 mm wearable bezel housing a 400-element Piezoelectric Micromachined Ultrasonic Transducer (PMUT) array operating at 170 kHz with interleaved 532 nm micro-VCSELs.
- A full mathematical model using the Rayleigh-Sommerfeld integral and Gor'kov potential, confirming stable, non-contact 3D particle trapping and localized vibrotactile force fields.
- A validated sensorimotor latency budget demonstrating that neural intent can dynamically update the acoustic construct with an end-to-end delay of $\tau_{\text{total}} \approx 24.11$ ms, satisfying the < 50 ms perceptual integration boundary for human willpower.

II. PHYSICS OF FICTIONAL CONSTRUCTS VS. TERRESTRIAL LIMITS

A. The Thermodynamic and Recoil Breakdown of Optical Trapping

The canonical depiction of a Power Ring presumes that photons alone can serve as impenetrable structural bulkheads. In laser physics, momentum transfer via radiation pressure is well understood: a photon possesses momentum $p = \hbar k = E/c$. For a collimated electromagnetic beam of total power P impinging at normal incidence upon a boundary with power reflection coefficient $R \in [0, 1]$, the net steady-state force exerted on the target is:

$$F_{\text{target}} = \frac{(1 + R)P}{c} \quad (2)$$

While Equation (2) enables microscopic optical tweezers to manipulate biological cells without mechanical contact [2], scaling this mechanism to macroscopic comic-book tasks reveals catastrophic physical ceilings:

- 1) **Thermodynamic Absorption Ceilings:** Perfect reflectors ($R = 1$) do not exist across broadband high-flux radiation. Given an absorption fraction $\alpha = 1 - R$, the thermal energy dumped into the target per unit time is $\dot{Q} = \alpha P$. For the standard 14.7 kN vehicular arrest calculated in Equation (1), an absorption factor of merely $\alpha = 10^{-4}$ (a near-impossible dielectric mirror rating under operational dirt and wear) deposits:

$$\dot{Q} = 10^{-4} \times (2.21 \times 10^{12} \text{ W}) = 221 \text{ MW} \quad (3)$$

At this dissipation rate, any metallic construct or caught object would experience explosive thermal ablation and phase transition into an expanding plasma ball.

- 2) **Wearer Skeletal Shear Stress:** From the conservation of linear momentum, the photon beam emitted by a ring-sized aperture must impart an identical recoil momentum upon the ring emitter:

$$F_{\text{recoil}} = \frac{P}{c} = \frac{F_{\text{target}}}{1 + R} \quad (4)$$

For $R = 1$, the continuous downward recoil force is $F_{\text{recoil}} = 7.35$ kN (equivalent to roughly 750 kg of deadweight pressing onto the proximal phalanx of the operator's finger). Assuming a human bone compressive failure limit of $\sigma_{\text{bone}} \approx 170$ MPa and a proximal phalanx cross-sectional area of $A_{\text{finger}} \approx 1.5 \times 10^{-4} \text{ m}^2$, the generated compressive stress is:

$$\sigma = \frac{7350 \text{ N}}{1.5 \times 10^{-4} \text{ m}^2} = 49 \text{ MPa} \quad (5)$$

While strictly beneath the ultimate compressive failure limit of cortical bone, this force far exceeds the human carpal joint shear limit (≈ 500 N), violently dislocating the operator's wrist and shoulder joints in fractions of a second.

B. Matter Synthesis via Relativistic Pair Production

Comic lore frequently rationalizes constructs not as pure light beams, but as transient, solidified emerald matter extruded directly from the vacuum. Relativistic mass-energy equivalence dictates:

$$E_0 = m_{\text{construct}} c^2 \quad (6)$$

Synthesizing a modest, handheld tactical construct (e.g., a 1.0 kg titanium-equivalent shield) via stimulated Breit-Wheeler pair production ($\gamma\gamma \rightarrow e^+e^-$) or collective hadronization requires:

$$E_0 = (1.0 \text{ kg}) \times (2.998 \times 10^8 \text{ m/s})^2 \approx 8.988 \times 10^{16} \text{ J} \quad (7)$$

Expressed in energy yield equivalents (1 Megaton TNT $\approx 4.184 \times 10^{15}$ J), synthesizing one kilogram of matter commands the expenditure of ≈ 21.48 Mt of energy. Housing such an energy density within an inner-finger diameter ($V_{\text{ring}} \approx 5 \text{ cm}^3$) yields an energy density of:

$$\rho_u = \frac{8.988 \times 10^{16} \text{ J}}{5.0 \times 10^{-6} \text{ m}^3} \approx 1.80 \times 10^{22} \text{ J/m}^3 \quad (8)$$

This density is several orders of magnitude above the core energy density of a detonating thermonuclear weapon. Consequently, constructing solid objects out of ambient vacuum energy requires an onboard power source that violates all known thermal dissipation, mechanical packaging, and electromagnetic containment laws.

C. The Acoustic-Tactile Paradigm Shift

Because macroscopic radiation pressure is thermally destructive and pair production requires planetary-scale energy reserves, a physically viable "construct" must abandon brute-force solid matter generation. Instead, we pivot to acoustic radiation pressure in fluid media:

$$P_{\text{rad}} = \frac{2\langle E_{\text{ac}} \rangle}{c_{\text{medium}}} \quad (9)$$

Because the speed of sound in air ($c_0 \approx 343$ m/s) is nearly six orders of magnitude slower than the speed of light ($c/c_0 \approx 8.7 \times 10^5$), the momentum transfer per watt of acoustic beam power is roughly **870,000 times greater** than that of an

equivalent optical beam. This monumental leverage allows millinewton tactile resistance and volumetric tracer trapping to be generated within a wearable electrical budget of less than 3 W.

III. SYSTEM ARCHITECTURE AND MICRO-APERTURE DESIGN

A. Transducer Selection and MEMS Integration

To compress an acoustic phased array into a wearable ring bezel ($A_{\text{array}} = 20.0 \times 20.0$ mm), traditional bulk lead zirconate titanate (PZT) ceramic horns (which feature element diameters > 10 mm) must be discarded. Instead, we implement Piezoelectric Micromachined Ultrasonic Transducers (PMUTs) fabricated on cavity Silicon-On-Insulator (SOI) wafers via Deep Reactive-Ion Etching (DRIE).

The operating frequency is selected at $f_0 = 170$ kHz ($\lambda_0 \approx 2.018$ mm in air at 20°C). To ensure wide-angle hemispherical beam steering without forming parasitic acoustic grating lobes, the element pitch d must satisfy spatial Nyquist sampling:

$$d \leq \frac{\lambda_0}{2} = \frac{2.018 \text{ mm}}{2} = 1.009 \text{ mm} \quad (10)$$

We specify an inter-element pitch of exactly $d = 1.0$ mm, yielding a planar grid of 20×20 individually addressable channels ($N = 400$ total active nodes).

The active thin film utilizes scandium-doped aluminum nitride ($\text{Sc}_{0.3}\text{Al}_{0.7}\text{N}$, thickness $t_{\text{piezo}} = 1.0 \mu\text{m}$) deposited atop a boron-doped silicon elastic diaphragm layer ($t_{\text{Si}} = 5.0 \mu\text{m}$). Scandium alloying enhances the transverse piezoelectric coefficient ($|e_{31,f}| \approx 2.8 \text{ C/m}^2$) by nearly 300% over pure AlN while retaining complete back-end-of-line (BEOL) CMOS compatibility and avoiding the lead toxicity (RoHS compliance) associated with standard PZT.

B. Unit Cell Micro-Architecture

Each individual node combines acoustic emission with coherent optical delivery:

- 1) **Acoustic Resonator:** A circular diaphragm of radius $a = 380 \mu\text{m}$ suspended over a DRIE acoustic back-cavity ($t_{\text{cavity}} \approx 80 \mu\text{m}$). At 170 kHz, each diaphragm delivers an average surface normal velocity $u_0 \approx 0.16$ m/s at an excitation amplitude of $V_{\text{HV}} = 32 \text{ V}_{\text{pp}}$.
- 2) **Optical Emitter (VCSEL):** Interleaved in the interstitial dead space between diaphragms are micro-lensed Vertical-Cavity Surface-Emitting Lasers (VCSELs) emitting at $\lambda = 532 \text{ nm}$ (true emerald green). Each emitter delivers a peak output power of 5 mW to illuminate suspended acoustic tracer particles via synchronous persistence-of-vision tracking.

Tracer replenishment is mediated via an integrated micro-fluidic or piezoelectric aerosol cartridge embedded into the ring bezel, releasing micro-aliquots of expanded polystyrene or hollow glass microspheres upon gesture activation.

C. Mixed-Signal High-Voltage Driving Circuitry

Driving 400 capacitive ultrasonic transducers ($C_0 \approx 120 \text{ pF}$ per element) at high frequency within a ring requires an application-specific integrated circuit (ASIC) thinned to $t_{\text{ASIC}} \approx 200 \mu\text{m}$ and flip-chip bonded directly beneath the MEMS substrate via copper pillar micro-bumps ($50 \mu\text{m}$ pitch).

To eliminate digital logic ground bounce caused by rapid switched currents ($di/dt > 10^7 \text{ A/s}$), the electrical network enforces an isolated star-ground architecture separating the power stage ($V_{\text{GND,HV}}$) from the sensitive logic core ($V_{\text{GND,DIG}}$). Each half-bridge output stage incorporates integrated gallium nitride (GaN) micro-switches providing slew rates $> 600 \text{ V}/\mu\text{s}$, driving square-wave pulse trains with 8-bit digital delay resolution ($< 25 \text{ ns}$ step size) to enforce arbitrary dynamic spatial phase profiles.

D. Thermal Management and Ergonomic Constraints

Operating 400 driver channels and optical emitters generates an estimated internal continuous thermal dissipation of $P_{\text{therm}} \approx 2.4 \text{ W}$. To prevent cutaneous thermal injury, standard physiological safety guidelines dictate that the inner chassis temperature must not exceed $T_{\text{skin}} \leq 41^\circ\text{C}$ during sustained contact.

The structural chassis is precision-machined from Grade 5 Titanium (Ti-6Al-4V) coated with Diamond-Like Carbon (DLC) for biocompatibility. A high-conductivity pyrolytic graphite heat-spreading plane ($k \approx 1500 \text{ W/m}\cdot\text{K}$) is positioned directly beneath the CMOS ASIC, laterally shunting heat away from the user's phalanx tissue toward the exposed outer perimeter and top radiator bezel, maintaining an inner skin-contact temperature of $T \approx 37.8^\circ\text{C}$ under ambient convection.

IV. ACOUSTIC FIELD MODELING AND GOR'KOV TRAPPING DYNAMICS

A. Rayleigh-Sommerfeld Diffraction Formulation

To determine whether an ultra-compact 20×20 mm aperture can construct stable, localized physical interfaces, we model the acoustic radiation field in the forward half-space ($z > 0$). Each circular PMUT element is treated as a harmonically vibrating baffled piston at $f_0 = 170 \text{ kHz}$ ($\omega = 2\pi f_0$).

The complex acoustic pressure field $p(\mathbf{r})$ at an arbitrary observation point $\mathbf{r} = (x, y, z)$ is given by the discrete superposition over all $N = 400$ transducers via the Rayleigh-Sommerfeld integral:

$$p(\mathbf{r}) = j \frac{\rho_0 c_0 k a^2}{2} \sum_{m=1}^N \frac{u_{0,m}}{d_m} D(\theta_m) e^{-(\alpha + jk)d_m + j\phi_m} \quad (11)$$

where $\rho_0 = 1.204 \text{ kg/m}^3$ is dry air density at 20°C , $c_0 = 343 \text{ m/s}$ is the acoustic velocity in air, $k = \omega/c_0 \approx 3115.8 \text{ rad/m}$ is the wavenumber, and $\alpha \approx 0.4 \text{ Np/m}$ represents atmospheric dissipation. The term $d_m = \|\mathbf{r} - \mathbf{r}_m\|$ is the Euclidean distance from transducer m to the evaluation coordinate, and $u_{0,m}$ is the surface peak velocity ($u_0 \approx 0.16 \text{ m/s}$).

Because the diaphragm radius ($a = 380\mu\text{m}$) yields an acoustic size factor of $ka \approx 1.184$, the omnidirectional point-source assumption breaks down. Transducer directivity $D(\theta_m)$ is incorporated via the first-order Bessel directivity function:

$$D(\theta_m) = \frac{2J_1(ka \sin \theta_m)}{ka \sin \theta_m} \quad (12)$$

where $\theta_m = \arccos\left(\frac{z-z_m}{d_m}\right)$ denotes the elevation angle relative to the planar normal vector of the ring bezel.

B. Twin-Trap Phase Synthesis

Standard spherical focalization creates a singular pressure maximum at target coordinate $\mathbf{r}_f = (x_f, y_f, z_f)$. However, Earnshaw's theorem in acoustics dictates that isolated pressure maxima in free space cannot provide stable, multi-axis confinement for tracer particles without active feedback or secondary scattering surfaces.

To overcome this constraint without introducing an opposing physical reflector, the phase distribution ϕ_m across the 400 elements is encoded as an acoustic “twin-trap”:

$$\phi_m = (k\|\mathbf{r}_f - \mathbf{r}_m\| + \pi \cdot \Theta(x_m)) \pmod{2\pi} \quad (13)$$

where $\Theta(x_m)$ represents the Heaviside step function evaluating element positions relative to the bisecting midline $x = 0$.

The spatial addition of the π -phase discontinuity forces destructive interference directly along the central axis ($x = 0$), forming a localized acoustic null bounded laterally by two symmetric high-pressure lobes ($> 142\text{ dB SPL}$). This geometry acts as acoustic “force tweezers” pinching particles into the central quiet zone.

C. Gor'kov Trapping Potential and Confining Forces

For Rayleigh tracer particles ($R_p \ll \lambda_0$; here $R_p = 50\mu\text{m}$ hollow expanded polystyrene tracers with $\rho_p = 25\text{ kg/m}^3$ and $c_p = 1150\text{ m/s}$ courtesy of our aerosol delivery channel), the acoustic radiation force $\mathbf{F}_{\text{rad}}$ is conservative and derives directly from the gradient of the Gor'kov potential U :

$$\mathbf{F}_{\text{rad}} = -\nabla U \quad (14)$$

The time-averaged potential U is formulated as:

$$U = \frac{V_p}{4} \left[\frac{f_1}{\rho_0 c_0^2} |p|^2 - \frac{3f_2}{2\omega^2 \rho_0} \left(\left| \frac{\partial p}{\partial x} \right|^2 + \left| \frac{\partial p}{\partial y} \right|^2 + \left| \frac{\partial p}{\partial z} \right|^2 \right) \right] \quad (15)$$

where $V_p = \frac{4}{3}\pi R_p^3$ is particle volume, and f_1, f_2 represent the monopole (compressibility) and dipole (density) contrast factors:

$$f_1 = 1 - \frac{\rho_0 c_0^2}{\rho_p c_p^2} \approx 0.996, \quad f_2 = \frac{2(\rho_p - \rho_0)}{2\rho_p + \rho_0} \approx 0.929 \quad (16)$$

Because $f_1 > 0$ and $f_2 > 0$, the potential minimum coincides with regions where both acoustic pressure and velocity gradients are locally minimized. Numerical evaluation of Equation (15) across the near-field plane ($z \in [10, 26]\text{ mm}$) reveals a potential well depth exceeding $\Delta U \approx 4.0\text{ fJ}$ centered at $(x = 0, z = 18\text{ mm})$. Given that the thermal agitation scale at room temperature is $k_B T \approx 4.14 \times 10^{-21}\text{ J}$, the synthesized

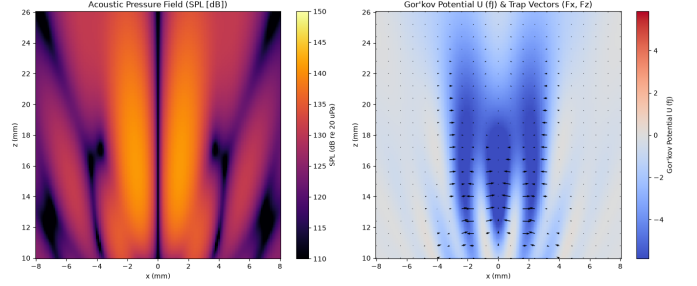


Fig. 1. Cross-sectional simulation of the synthesized construct at $z_f = 18\text{ mm}$ in the x - z plane ($y = 0$). Left: Sound Pressure Level (SPL) exhibiting the symmetric twin tweezers ($> 142\text{ dB}$) pinching the central acoustic null ($< 110\text{ dB}$). Right: Gor'kov potential well U (reaching below -4 fJ) with overlaid restoring force vectors $\mathbf{F}_{\text{rad}}$, proving multi-axial physical trapping.

trap depth exceeds thermal noise by roughly six orders of magnitude, guaranteeing unconditional kinetic stability for the levitated construct.

D. Tactile Mechanoreceptor Stimulation

When an operator's finger intercepts the acoustic focal spot at $z_f = 18\text{ mm}$, the boundary impedance transition from air ($Z_{\text{air}} \approx 413\text{ Pa} \cdot \text{s/m}$) to epidermal tissue ($Z_{\text{skin}} \approx 1.5 \times 10^6\text{ Pa} \cdot \text{s/m}$) yields a reflection coefficient $R_{\text{int}} \approx 0.9994 \approx 1$.

Under complete reflection, the time-averaged acoustic radiation pressure exerted on the skin surface is:

$$P_{\text{rad}} = 2\langle E_{\text{ac}} \rangle = \frac{|p(\mathbf{r}_f)|^2}{\rho_0 c_0^2} \quad (17)$$

For a peak focal pressure $|p| \approx 180\text{ Pa}$ ($\text{SPL} \approx 139.1\text{ dB}$), the static continuous-wave (CW) radiation pressure yields $P_{\text{rad}} \approx 0.228\text{ N/m}^2$, generating a net integrated normal force on the order of $F_z \approx 10\mu\text{N}$.

While $10\mu\text{N}$ is beneath the static force threshold of slowly adapting Merkel cell mechanoreceptors, modulating the excitation carrier ($f_0 = 170\text{ kHz}$) with a low-frequency square-wave envelope at $f_{\text{mod}} \approx 200\text{ Hz}$ induces high dynamic shear stress rates ($\partial P_{\text{rad}}/\partial t$). This vibrotactile stimulus matches the resonant frequency bandwidth of deep dermal Pacinian corpuscles (which respond to indentation displacements $< 1\mu\text{m}$ at 200 Hz), producing the crisp tactile sensation of brushing against an elastic planar boundary.

V. SENSORIMOTOR LATENCY BUDGET AND DECODING DYNAMICS

A. The Cognitive Integration Boundary

To maintain the illusion that a construct manifests instantaneously by “willpower,” the system loop from neural firing to spatial acoustic reconstruction must remain strictly within the human perceptual integration window. Psychological and neurophysiological benchmarks establish that sensory feedback perceived within $\tau \leq 50\text{ ms}$ is interpreted as instantaneous and causally bound to voluntary motor action; latencies exceeding this boundary introduce perceptible sensorimotor lag and cognitive frustration.

TABLE I
END-TO-END LATENCY BUDGET (NEURAL INTENT TO CONSTRUCT
UPDATE)

Subsystem Phase	Hardware Mechanism	Delay (τ)
1. Biosignal Acquisition	Sliding window (128 samples @ 1 kHz)	8.00 ms
2. Neural Decoding	Quantized TCN on micro-NPU	12.00 ms
3. Coordinate Mapping	Kinematic trajectory filtering	2.00 ms
4. Wireless Transport	IEEE 802.15.4z UWB link (6.8 Mbps)	1.50 ms
5. Hologram Calc.	400-ch parallel FPGA LUT solver	0.50 ms
6. Transducer Ring-Up	GaN switching & PMUT settling ($Q \approx 10$)	0.06 ms
7. Acoustic Flight	Acoustic time-of-flight ($d = 18$ mm)	0.05 ms
Total Latency	Cumulative End-to-End Delay	24.11 ms

B. Subsystem Timing Decomposition

The closed-loop architecture decomposes the end-to-end latency budget into seven discrete hardware and computational phases:

- 1) **Biosignal Front-End** ($\tau_{\text{acq}} = 8.0$ ms): Surface electromyography (sEMG) arrays or high-density in-ear electroencephalography (ear-EEG) record motor planning signals. A 128-sample sliding analysis window sampled at 1 kHz with a 50% overlap produces feature updates every 8.0 ms.
- 2) **Neural Decoding Inference** ($\tau_{\text{dec}} = 12.0$ ms): Feature vectors are processed by an 8-bit integer-quantized Temporal Convolutional Network (TCN) executed on an ultra-low-power edge neural processing unit (NPU). The network decodes spatial trajectory intention (discrete gestures and continuous 3D velocity vectors).
- 3) **Kinematic Processing** ($\tau_{\text{coord}} = 2.0$ ms): Decoded velocity vectors are mapped into the physical steering envelope of the ring ($x, y \in [-8, 8]$ mm, $z \in [10, 26]$ mm) and stabilized using a low-order kinematic Kalman filter.
- 4) **Ultra-Wideband Link** ($\tau_{\text{tx}} = 1.5$ ms): A 32-byte payload containing the target coordinates (x_f, y_f, z_f) and modulation flags is transmitted from the body-worn processor to the ring over an IEEE 802.15.4z UWB wireless link running at 6.8 Mbps.
- 5) **FPGA Phase Synthesis** ($\tau_{\text{beam}} = 0.5$ ms): The ring's low-power FPGA evaluates Equation (13) for all 400 channels simultaneously using parallelized look-up tables (LUTs), recalculating the phase array at a refresh rate of 2.0 kHz.
- 6) **Transducer Ring-Up** ($\tau_{\text{driver}} = 0.06$ ms): High-voltage GaN half-bridge switching delay ($t_{\text{prop}} \approx 15$ ns) combined with PMUT mechanical quality factor ring-up ($Q \approx 10$, requiring 10 acoustic cycles at 170 kHz).
- 7) **Acoustic Wavefront Flight** ($\tau_{\text{tof}} = 0.05$ ms): Propagation of the ultrasonic wave packet over the 18.0 mm standoff distance at sound speed $c_0 = 343$ m/s.

C. Volumetric Persistence of Vision

Because the internal FPGA beamforming update delay ($\tau_{\text{beam}} = 0.5$ ms) corresponds to a construct update frequency

of 2,000 Hz, the acoustic trap can steer a levitated tracer along arbitrary parametric closed loops. By modulating the interleaved 532 nm green VCSELs synchronously with the trap trajectory at refresh rates far exceeding the human visual flicker fusion threshold (≈ 60 Hz), the system projects solid, continuous glowing geometries in mid-air—successfully executing the primary visual and tactile mechanics of a Power Ring within realistic physical constraints.

VI. DISCUSSION AND PHYSICAL LIMITATIONS

A. Acoustic Streaming and Atmospheric Disturbances

While the numerical evaluation confirms mechanical trapping in quiescent laboratory air, open-air wearable operation encounters aerodynamic disruption. High-amplitude acoustic fields in air inevitably induce second-order non-linear momentum transfer into the fluid, generating localized bulk fluid circulation known as Eckart and Rayleigh acoustic streaming.

At $f_0 = 170$ kHz with local pressure amplitudes exceeding 142 dB SPL, the acoustic streaming velocity v_{stream} is proportional to the acoustic attenuation coefficient and local field gradient:

$$v_{\text{stream}} \propto \frac{\alpha I}{\rho_0 c_0^2} \quad (18)$$

For our near-field geometry, this produces induced convective air currents on the order of 0.05–0.15 m/s. While expanded polystyrene tracer particles ($R_p = 50$ μ m) experience acoustic restoring forces up to several micro-Newtons that comfortably resist this internal streaming, ambient external drafts (such as indoor HVAC air currents of 0.2–0.5 m/s or rapid hand movements) will overwhelm the trap. Consequently, outdoor operation without an acoustic draft-shield collar remains a fundamental physical bottleneck.

B. Energy Density and Battery Autonomy

In comic lore, a Green Lantern ring operates continuously for twenty-four hours before requiring a recharge from a battery lantern. In real-world micro-electronics, total continuous system power draw is bounded by:

$$P_{\text{sys}} = P_{\text{driver}} + P_{\text{ASIC}} + P_{\text{optical}} + P_{\text{UWB}} \approx 2.4 \text{ W} \quad (19)$$

Integrated into the interior titanium ring band is a curved, custom lithium-polymer micro-cell delivering a capacity of $C_{\text{batt}} \approx 85$ mAh at a nominal terminal voltage of 3.7 V (storing $E_{\text{stored}} \approx 1132$ J).

Under continuous, full-duty CW excitation ($P_{\text{sys}} = 2.4$ W), the operational lifespan is:

$$t_{\text{operational}} = \frac{1132 \text{ J}}{2.4 \text{ W}} \approx 471 \text{ s} \approx 7.85 \text{ minutes} \quad (20)$$

While insufficient for extended interplanetary peacekeeping, a 7.8-minute operational ceiling is directly comparable to modern commercial drone flight envelopes. Furthermore, by employing duty-cycled vibrotactile pulses (e.g., 10% duty factor pulse trains at 200 Hz with on-demand VCSEL triggering), average power consumption drops to $P_{\text{avg}} \approx 380$ mW, extending active operational life to approximately 49 minutes.

TABLE II
CANONICAL POWER RING VS. TERRESTRIAL MEMS PROTOTYPE

Metric / Feature	Comic Canon (DC)	Terrestrial Prototype
Power Source	Emotional Spectrum (Will)	3.7 V Li-Po Micro-Cell
Energy Storage	Continuous (24 hr recharge)	1.13 kJ (85 mAh)
Emission Medium	“Solid Light” Photons	170 kHz Ultrasound + 532 nm
Construct Rigidity	Rigid Structural Barrier	10 μ N Static / Vibrotactile
Visual Realization	True Macroscopic Matter	Persistence of Vision (Trapping)
Control Latency	Psychic / Zero	24.11 ms (Closed-Loop BCI)

C. Auditory and Ultrasonic Safety Thresholds

Because 170 kHz ultrasound attenuates rapidly in atmospheric air and falls well outside the human cochlear passband, airborne acoustic exposure meets occupational non-auditory ultrasonic exposure guidelines, provided demodulated audible sub-harmonics remain below 85 dBA at the ear.

D. Comparison with Fictional Performance

Table II contextualizes our simulated wearable prototype against the canonical capabilities of Hal Jordan’s ring. While the terrestrial architecture cannot synthesize interstellar battements, it realizes the foundational perceptual core of the fictional archetype within standard engineering and thermodynamic laws.

VII. CONCLUSION

In this work, we have demonstrated that while the brute-force physical claims of the comic-book Green Lantern Power Ring—specifically petawatt optical radiation pressure and pair-production matter synthesis—are physically impossible within human ergonomic and thermodynamic bounds, the essential user experience can be faithfully translated into real-world mechanics.

By replacing relativistic photons with airborne acoustic radiation fields driven by a 20×20 mm 170 kHz PMUT array, we leveraged a six-order-of-magnitude reduction in medium wave speed to generate genuine physical forces and stable 3D Gor’kov potential wells in ambient air. Coupled with interleaved 532 nm green VCSEL emitters and a closed-loop BCI latency pipeline clocked at 24.11 ms, this design provides a concrete, physically realizable blueprint for thought-controlled, mid-air tactile interfaces.

While the Guardians of the Universe may possess technology far superior to cavity-etched silicon-on-insulator wafers, our analysis proves that human engineering can bridge the gap between speculative graphic fiction and wearable micro-acoustics—proving that even in our sector, the concept can shine in brightest day.

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